PREPARATION OF DATA ON SW CERAMIC PRODUCTION FOR PATH GENERATION

The base data for the simulation of the control group sim_SW stems from De Roever (2004). De Roever (2004) does not provide quantitative data on complete specific *chaînes opératoires* for individual vessels but instead provides a single, abstracted discussion of technical traces on ceramics from S3. This supplement outlines the re-interpretation of these traces to form the random seed for sim_SW (see Main text). The description follows the terminology and stages of the ceramic *chaîne opératoire* described in Roux (2019).

Interpretation of ceramic chaînes opératoires

The fashioning of the base is the first step in SW ceramic production (De Roever 2004, 48). This step could be reconstructed for 28 vessel bases. Two procedures are described based on the breakage pattern of the base and the orientation of voids and particles in vessel bases (cf. Roux 2019, 160-5). The first procedure (n=11) involves coiling of the vessel base with the spiral procedure (cf. Roux 2019, 55). The second procedure (n=17) involves modelling, presumably by pinching (cf. Roux 2019, 60; 168) of a small ball or cone-like clay mass. Regardless of this procedure, the first coils of the vessel wall are then affixed adjacent to the base.

De Roever (2004, 47-8) discusses the following traces for the fashioning of the vessel wall: a horizontal breakage pattern, parallel horizontal orientation of elongated temper particles on breaks, and the appearance of hollow, straight, and bevelled break profiles. Multiple break profiles are said to appear within the same vessel. For example, break profiles may feature internal bevels on the lower body and external bevels in the upper body (eg, De Roever 2004, Fig. 12b). These traces indicate the use of the coiling technique following the segment or ring procedure, and joining by pinching or spreading which alternate joining methods depending on the position of the coil in the vessel wall (cf. Roux 2019, 55-8; 160-5).

De Roever (2004) describes three preforming techniques: preforming by pressure with hands, scraping, and the application of separate elements. The indicators for preforming by pressure with hands are elongated, finger-sized hollows which appear on the lower interior surfaces and around the rim of the vessel (De Roever 2004 47-8; cf. Roux 2019, 64-6; 174-6). De Roever (2004, 49; 198) also discusses and depicts traces of scraping: namely bands of broad, deep striations with thickened edges and an irregular microrelief at the base (cf. Roux 2019; 64-6; 175). The traces of this latter operation are said to only appear if no subsequent operation has been applied during the production process, indicating a position in the ceramic

chaîne opératorie prior to the finishing of the vessel wall. De Roever (2004, 49) states traces of scraping do not occur on all vessels, but provides no quantitative information about the frequency of this technique. To obtain this frequency, this study tallied the number of depicted vessels from S3 (see De Roever 2004, Fig. 9-20; Appendix) of which the descriptions mention traces of scraping. This results in a total of 9 out of 121 depicted vessels. The last preforming techniques is the application of separate elements. De Roever (2004, 58; Fig. 19o-q) mentions 3 vessels (out of 484) feature separately applied elements in the shape of perforated handles. These handles break away from the vessel wall separately and feature finger-sized depressions both on the handle and on the surrounding surface. In addition, the perforation features ribbed or threaded striations, indicating it was performed while the clay of the handle was in a wet state (cf. Roux 2019 90; 152-3). Therefore, the handles were applied through pressure alone onto the vessel wall while the clay of the vessel wall and handle was in the wet state.

The technique applied during the finishing stage of the ceramic production process is wet smoothing by hands with some added water (De Roever 2004, 49). The photographs (De Roever 2004, Appendix) of ceramics indeed show sherds with irregular microtopography, protruding particles, matt aspect, and striations with threaded or ribbed edges, corroborating this interpretation (cf. Roux 2019, 93; 196-7).

The next stage in the SW ceramic production process is the application of decorative techniques. Three decorative techniques are present, going by the descriptions, drawings, and photographs. De Roever (2004) does not report consistent frequencies of these techniques relative to vessel counts, hence the frequencies reported here are based on the descriptions of depicted vessels (see above).

The most common decorative technique (61 out of 121 depicted vessels) is excision. De Roever (2004, 56-7, Photograph 2) describes teardrop-shaped depressions, which deformed the vessel wall and feature irregular microrelief and threaded or ribbed striation at the base. These depressions would have been made by inserting a hollow object at an oblique or perpendicular angle into the vessel wall and extracting part of the vessel wall. Therefore, they are classified as excisions on wet clay (cf. Roux 2019, 108; 149-54). The second most common technique (10 out of 121 depicted vessels) is simple impression, which was performed by pressing a spatula or finger into the wet clay, leaving marks of these tools (De Roever 2004, 57; eg, Fig. 18i-j; cf. Roux 2019, 104-6). The third most common decorative technique is simple incision, which occurs in 7 out of 121 depicted vessels and is argued to have been applied to wet clay (De Roever 2004, 57; eg, Fig. 18w; cf. Roux 2019, 107). The

final decorative technique is decorative modelling. Traces of this technique are impressions made with finger tops into the clay wall, after which the clay was pinched and deformed (eg, De Roever 2004, Photograph 5; Fig. 20; cf. Roux 2019, 109). Traces of this technique occur in 6 out of 121 depicted vessels (De Roever 2004, Appendix).

Most SW ceramics feature only one decorative technique, but De Roever (2004) depicts three with multiple decorative techniques. The first of these vessels features a combination of simple incisions and simple impressions, with the drawings suggesting traces of the latter technique cut across traces of the former technique (De Roever 2004, Fig. 20i). Hence this order of techniques is present in the dataset. The second vessel features a combination of excisions and simple impression (De Roever 2004, Fig. 10h), and the third vessel simple incisions plus simple impressions (De Roever 2004, Fig. 10c). For these last two vessels, the drawings and documentation do not offer clues as to the order of application for these techniques. Therefore, it is assumed here that the decoration higher up on the vessel was applied prior to the decoration further down.

The surface treatment applied to SW vessels consists exclusively of burnishing. De Roever (2004, 49) describes the following traces of burnishing: surfaces with a compact microtopography, glossy aspect, and bundled indistinct striations. These traces are consistent with burnishing applied to leather-hard clay (cf. Roux 2019, 96-8; 201; Lepère 2014). The frequency of these traces is 23 out of 121 depicted and described vessels (cf. De Roever 2004, Appendix). The edges of the depressions left by decorative techniques (see above) were reworked during surface treatment (De Roever 2004, 56), indicating surface treatment was performed after the application of surface treatment techniques.

De Roever (2004) does not discuss further operations between surface treatment, drying to dry consistency, and firing. Therefore, the order of these operations is assumed to be fixed: surface treatment (if present), drying to dry consistency, and firing.

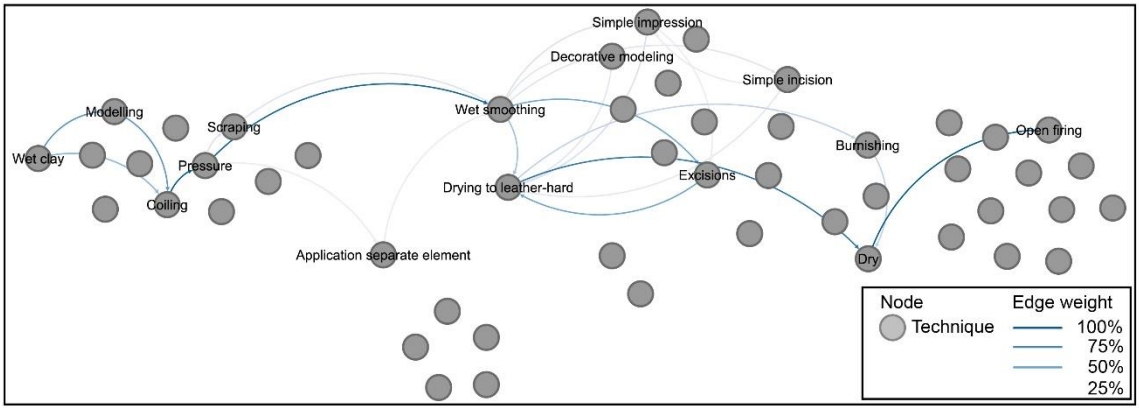
The firing process of the SW vessels resulted in a mottled exterior and a firing core with brightly coloured interior and exterior surfaces which sharply transition into a dark grey core (De Roever 2004, 49). These traces indicate the vessel may have been fired in an upright position under conditions which did not allow for the combustion of carbon. The oxidising colours on the exteriors indicate an exposure to cool, oxygen-rich air towards the end of the firing process (cf. Roux 2019, 207-9; Rye 1981, 114-8). Re-firing and X-Ray Diffraction analysis indicate firing temperatures generally below 700°C (De Roever 2004, 118). In general, these traces are consistent with open firing (Thér *et al.* 2019).

Conversion to weighted subgraph

Based on the above findings, the following edge weight distribution is adopted for the SW random seed (see Table 1). A network representation of this random seed is shown in Supplementary Figure 1. Only the combinations of techniques which are possible within the observed data are shown in these overviews. Note that the relative frequency of separately applied handles has been set to 1%, whereas the actual frequency is approx. 0.2% (see above). This value has been rounded ups because the random generation process is currently not compatible with decimal values (cf. Kroon 2024, Appendix F).

Source node	**Target node**	**Weight**
Wet clay	Coiling	39
Wet clay	Modelling	61
Coiling	Preforming by pressure	100
Modelling	Coiling	61
Drying to leather-hard	Drying to dry	81
Drying to leather-hard	Burnishing	19
Scraping	Wet smoothing	7
Preforming by pressure	Scraping	7
Preforming by pressure	Application of separate elements	1
Preforming by pressure	Wet smoothing	92
Application of separate elements	Wet smoothing	1
Wet smoothing	Drying to leather-hard	31
Wet smoothing	Simple impressions	8
Wet smoothing	Simple incisions	6
Wet smoothing	Excisions	50
Wet smoothing	Excisions	5
Drying to dry	Open firing	100
Burnishing	Drying to dry	19
Simple impressions	Drying to leather-hard	9
Simple impressions	Excisions	1
Simple incisions	Drying to leather-hard	4
Simple incisions	Simple impressions	2
Excisions	Drying to leather-hard	51
Excisions	Drying to leather-hard	5

Table 1: Edge weight distribution for the random seed of SW ceramic production, based on data from De Roever (2004). The edge weights are the relative frequencies of techniques rounded to integers to facilitate guided random path generation (cf. Kroon 2024).



Supplementary figure 1: Weighted subgraph of Fig. 6 with the random seed for sim_SW. The nodes are techniques; directed edges represent observed techniques which can form a sequence (cf. Roux 2019). For edge weights, see Table 1. Lay-out with the ForceAtlas 2 algorithm in Gephi (Bastian *et al.* 2009; Jacomy *et al.* 2014).

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